(iii) $(A \vee B) \wedge C = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$ (iv) $(A \wedge B) \vee C = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

10. (i) It is commutative and associative.
    (ii) Identity and Inverse is exist.

Exercise 12.2

1. (i) $\neg p$: Jupiter is not a planet (ii) $p \wedge \neg q$: Jupiter is a planet and India is not an Island.
   (iii) $\neg p \vee q$: Jupiter is not a planet or India is an Island.
   (iv) $p \rightarrow \neg q$: If Jupiter is a planet then India is not an Island.
   (v) $p \leftrightarrow q$ Jupiter is a planet if and only if India is an Island.
2. (i) $\neg p \wedge q$ (ii) $p \vee \neg q$ (iii) $p \wedge q$ (iv) $\neg p$
3. (i) $p \rightarrow q$ is $\mathbb{T}$ (ii) $p \vee q$ is $\mathbb{F}$ (iii) $\neg p \vee q$ is $\mathbb{T}$ (iv) $p \wedge q$ is $\mathbb{F}$
4. (i), (iii) and (iv) are propositions
5. (i) **Converse:** If x and y are numbers such that $x^2 = y^2$ then $x = y$.
   Inverse: If x and y are numbers such that $x \neq y$ then $x^2 \neq y^2$.
   Contra positive: If x and y are numbers such that $x^2 \neq y^2$ then $x \neq y$.
   (ii) **Converse:** If a quadrilateral is a rectangle then it is a square.
   Inverse: If a quadrilateral is not a square then it is not a rectangle.
   Contrapositive: If a quadrilateral is not a rectangle then it is not a square.
6. (i) Truth table for $\neg p \wedge \neg q$

p	q	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	F	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

(ii) Truth table for $\neg(\neg p \wedge \neg q)$

p	q	$\neg q$	$p \wedge \neg q$	$\neg(\neg p \wedge \neg q)$
T	T	F	F	T
T	F	T	T	F
F	T	F	F	T
F	F	T	F	T

(iii) Truth table for $(p \vee q) \vee \neg q$

p	q	$\neg q$	$p \vee q$	$(p \vee q) \vee \neg q$
T	T	F	T	T
T	F	T	T	T
F	T	F	T	T
F	F	T	F	T

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